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Computer Science Academy

Description

Describe your program. What do you envision your app will do?

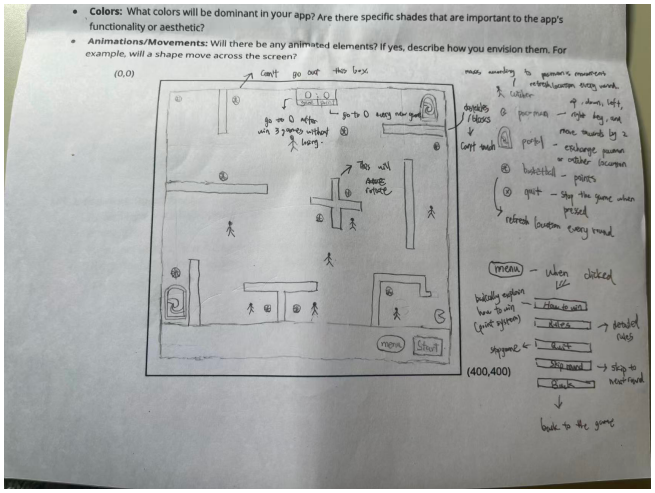
Note: At this step, do not be concerned about how to do it, just focus on what you envision.

I want to use the knowledge I learned in the past four units to create a structure of a pac-man game. This structure should include all the basic buttons, movement of the pac-man, as well as a system of where each catcher, block, and pellet should be located at. I am not going to worry about some in-depth eating or rewarding part since we have not covered that yet, but I do want to code a simple game outline. For the buttons, I want to have start, menu, up, down, left, right, and a cancel button. The up, down, left, right button should actively change its opacity according to the movement of the pac-man. Meanwhile, a restriction should be set between the blocks and the pac-man, limiting it from reaching to pellets easily. The catcher should not have function at this stage and so does the pellet eating system.

Canvas Design

Sketch your canvas. Label any animation or user interaction.

- **Colors:** What colors will be dominant in your app? Are there specific shades that are important to the app's functionality or aesthetic?
- **Animations/Movements:** Will there be any animated elements? If yes, describe how you envision them. For example, will a shape move across the screen?



Code Design

Plan your code. List here the coding concepts and properties you plan to use to accomplish your goals, and why you plan to use them. These should incorporate at least two new concepts from this unit. Briefly describe what purpose each element will serve in your project. Aim for at least four to six properties.

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Concept	Purpose
if-elif	To move the pac-man, to change the opacity of pac-man controller (fake), and to manipulate the menu
OnKeyPress	To move the pac-man using “up”, “down”, “left”, and “right” key
Shape Method - hits shape	This bounces the pac-man back whenever it hits the wall (boundary of the game machine) or hits the blocks
Add Shapes Randomly	To add a fixed number of catchers and pellets each time randomly at a random location with a restricted domain and range

Creative Task Reflection

After completing your creative task use this form to reflect on your work.

Difficulties and Opportunities

For this section, go in-depth into either a difficulty you faced or an opportunity you saw to improve or add to your code.

If you Faced a Difficulty:

- Describe the Problem: What specific issue did you encounter?
- Attempted Solutions: What did you try in order to solve the problem?
- Final Solution: What was the final solution that fixed the issue?

If you Found an Opportunity:

- Reason for the Addition/Change: What made you consider this opportunity? Was it a challenge or an aha-moment?
- Impact on Design: How did seizing this opportunity impact other aspects of your design?
- Design Trade-offs: What aspects of your original design did you have to adjust or remove to incorporate this new feature?

One difficulty I remember clearly is when I was coding the bouncing back function of the block and boundary. The process of coding the bouncing function for the block went smoothly, but the boundary bouncing function went completely. Since block is a shape inside the “game machine”, but the boundary is the border of the game machine, in which the game machine itself is made by a rectangle. I originally tried to solve this by setting domain and range of pac-man’s movement so that it doesn’t reach any coordinate over the border. However, my code did not work out and it is also not really learned. Therefore, I came up with a better solution of using 4 more rectangles to surround the border with fill None. With these 4 rectangles, I can include them with the block and other features into a collective group and code the bouncing function for the entire group. What I learned from solving this issue is that

people are often scared of not knowing enough when they already had the ability to solve problems critically when they think outside the box

Finished Project Code Outline

List **only one instance** of each important coding concept used in your project, along with the corresponding line number, and describe how it contributes to the overall function of your program. **Highlight the aspect you are most proud of.** For each concept, also include a brief definition to demonstrate your understanding. By focusing on one instance of each concept, aim to provide more depth and clarity in your responses.

Concept	Line number(s)	Definition	Purpose
if-elif	87-89	This is a conditional statement that executes a series of code as long as a certain condition is met. And elif is basically just the secondary code if the first condition is not met, it supports a second condition	I used this feature to set a condition where my pac-man touched the boundaries and blocks in order for it to bounce back into different directions according to the original directions of my pac-man
OnKeyPress	95-113	This is a execution of a series of code when you press a key on the key board	This code is a general way how you code the movement of your game's character using up, down, left, and right. And I used exactly this feature to code my pac-man's movement with changes in the opacity of the controller to indicate the OnKeyPress
Shape Method - hits shape & On MousePress	147-161	This is a execution of a series of code when the mouse is pressed on another shape	I use this feature to help open my menu and navigate into a menu page consist of 5 choices for display, including a go back button to return to the game
Add Shapes Randomly	52-62	This function add certain shape(s) at a fixed number within a restricted boundary, which refresh location every round	I use this feature to code my pellets and catcher so they can appear at different locations to make the game more entertaining

Future goals

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Reflect on your entire design and coding process, from conceptualization to implementation. In this section, consider not only what you would change in the final product but also how you would approach the entire project differently.

- **Code Changes:** What specific changes would you make to your code if you had more time or if you were to start from scratch? Would you add new features or refine existing ones?
- **Process Adjustments:** Did you stick to the original plan, or were there deviations? Were these changes beneficial or did they lead to complications? How could your planning or approach be improved?
- **Resource Utilization:** Did you make the best use of available resources, like class notes, tutorials, or peer help? What additional resources could have been beneficial?
- **Time Management:** Reflect on how effectively you managed your time. What would you do differently to ensure a smoother process?

By considering these aspects, you will not only think about 'what' you would change but also 'how' and 'why,' which can be incredibly enlightening for future projects.

In my opinion, I would certainly add a little bit more feature to this game since it can not be really count as a “game”, but more of a structure that I can build on in the future. Some new features would be eating those pellet, adding a rulebook, adding the catchers’ movements, as well as the rotation of some blocks. My second improvement should certainly be time management as well as the planning of my code. It is important for me to oversee what I need to code and when i should reach out to teacher. I did not know the greatness of reaching out and asking deep questions until last class, which I used sandbox to receive an incredible nice code template to be used for my project. But by then I was already busy with other activities like CISSMUN and didn’t have time to reach out for more. Thus, reaching out earlier and prepare good questions is certainly one of the things I need to improve on the most. Lastly, I believe using some additional research could also contribute to the success of this project, since python is used in many place and it would often give you unique ideas that you’ve never thought of in the past (which I did eventually and realize how powerful it is to just google something out!)